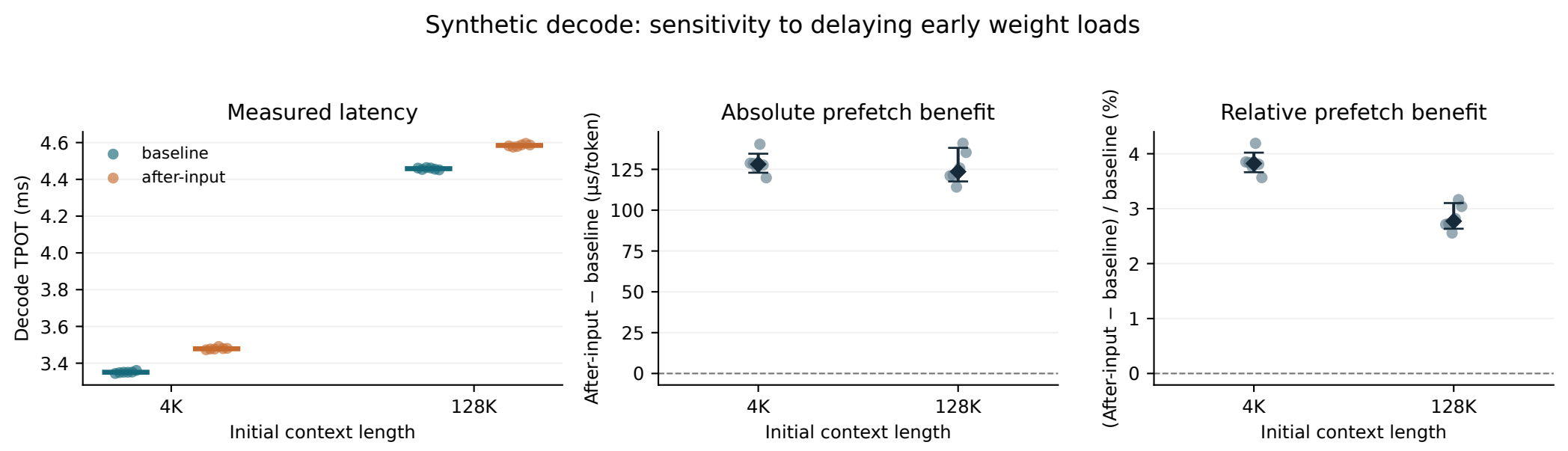
Synthetic decode: sensitivity to delaying early weight loads
Measured latency
Decode TPOT (ms)
4.6
4.4
4.2
4.0
3.8
3.6
3.4
baseline
after-input
4K
128K
Initial context length
Absolute prefetch benefit
After-input − baseline (µs/token)
125
100
75
50
25
0
4K
128K
Initial context length
Relative prefetch benefit
(After-input − baseline) / baseline (%)
4
3
2
1
0
4K
128K
Initial context length
H100 · BS1 · synthetic weights & KV · 6 paired repeats · 511 decode steps after process-local warmup
Dots: all repetitions/pairs. Diamonds: median with 95% paired-bootstrap interval. Positive benefit favors early prefetch; no HBM mechanism is inferred.